

On dividing 999999 by 840, the remainder will be 399.
Therefore, the greatest six digit number divisible by 840 is

$$999999 - 399 = \mathbf{999600}$$

Since $5 - 3 = 6 - 4 = 7 - 5 = 8 - 6 = 2$, the remainder in each case is less than the divisor by 2.

$\therefore$ The required number = $999600 - 2 = \mathbf{999598}$

EXAMPLE 37 What is the least multiple of 11, which when divided by 3, 4, 5, 6 and 7, leaves the remainder 2, 3, 4, 5 and 6 respectively?

Solution The LCM of 3, 4, 5, 6 and 7 = 420.

Since $3 - 2 = 4 - 3 = 5 - 4 = 6 - 5 = 7 - 6 = 1$, if you add one to the required number, the resulting number will be divisible by the given divisors.

But we require the number as a multiple of 11.

Assume the number to be $420 \times n - 1$

i.e. $(38 \times 11 + 2)n - 1$

$\therefore 38 \times 11n$ is divisible by 11, the number to be checked for divisibility by 11 is $2n - 1$.

The least value of $2n - 1$ to be divisible by 11 is 11

i.e. $2n - 1 = 11 \Rightarrow n = \frac{1 + 11}{2} = 6$

$\therefore$ The required number = $420 \times 6 - 1$
 $= \mathbf{2519}$

EXAMPLE 38 What is the least number which when divided by 12, 16 and 18 leaves 9 as remainder in each case, but when divided by 27 leaves no remainder.

Solution The number which is divisible by all the three given numbers is of the form $144k$, when k is an integer.

Here, the number leaves 9 as remainder on division

$\therefore$ Required number = $144k + 9$.

Given $(144k + 9) \div 27 = 0$

If $k = 5$, value of number will be equal to $144 \times 5 + 9 = 729$

Hence, required number = $\mathbf{729}$



EXERCISES

- The sum of two numbers is 468 and their HCF is 39. How many pairs of such numbers can be formed?
(a) 11 (b) 22 (c) 33 (d) 36 (e) None of these
- Find the sum of the numbers between 500 and 700 such that when they are divided by 6, 8 and 12
(i) it leaves no remainder
(a) 3200 (b) 4600 (c) 5300 (d) 5400 (e) None of these
(ii) it leaves 5 as remainder in each case
(a) 4600 (b) 4447 (c) 4744 (d) 4848 (e) None of these

3. Is it possible to divide 1,000 into two parts such that their HCF is 15?
(a) Yes, it is possible (b) No, it is sometimes possible
(c) Yes, it may be possible (d) Never possible
4. In a school, 442 boys and 374 girls have been divided into the largest possible equal classes, so that each class of boys numbers the same as each class of girls. What is the number of classes?
(a) 16 (b) 18 (c) 20 (d) 24 (e) None of these
5. Find the greatest number of 4 digits and the least number of 5 digits that have 196 as their HCF.
(a) 9998, 10184 (b) 9996, 10192 (c) 9994, 10194 (d) 9998, 10196 (e) None of these
6. The product of two numbers is 4212 and their HCF is 9. Find the possible pair of numbers.
(a) 32, 121 (b) 36, 117 (c) 42, 114 (d) 45, 110 (e) None of these
7. The HCF and LCM of two numbers are 36 and 756 respectively. Find the possible pair of numbers.
(a) 72, 378 (b) 108, 252 (c) 144, 658 (d) 180, 622 (e) None of these
8. Find the maximum possible length which can be used to measure exactly the lengths 6 m, 4 m 75 cm, and 10 m 25 cm.
(a) 20 cm (b) 25 cm (c) 30 cm (d) 35 cm (e) None of these
9. The LCM of two numbers is 1008 and their HCF is 36. If one of the two numbers is 144, find the other number.
(a) 288 (b) 252 (c) 576 (d) 504 (e) None of these
10. Three bells ring together and they will ring at intervals of 6 seconds, 7 seconds and 8 seconds respectively. After what interval will they again ring together?
(a) 178 seconds (b) 174 seconds (c) 168 seconds (d) 164 seconds (e) None of these
11. Find the least number of square tiles required to pave the ceiling of a hall 17 m 55 cm long and 8 m 10 cm broad.
(a) 78 (b) 64 (c) 84 (d) 92 (e) None of these
12. Find the smallest whole number which is exactly divisible by 6, 8, 15, 18 and 24.
(a) 120 (b) 240 (c) 740 (d) 360 (e) None of these
13. Find the highest common factor of 36 and 84. [RRB]
(a) 4 (b) 6 (c) 12 (d) 18
14. Find the least common multiple of 24, 36 and 40. [RRB]
(a) 120 (b) 240 (c) 360 (d) 480
15. The ratio of two numbers is 3 : 4 and their HCF is 4. Their LCM is [SSC]
(a) 12 (b) 16 (c) 24 (d) 48
16. The product of two numbers is 4107. If the HCF of these numbers is 37, then the greater number is [SSC]
(a) 101 (b) 107 (c) 111 (d) 185
17. The HCF and LCM of two numbers are 21 and 84 respectively. If the ratio of the two numbers is 1 : 4, then the larger of the two numbers is [MAT]
(a) 12 (b) 48 (c) 84 (d) 108
18. The HCF of two numbers is 8. Which one of the following can never be their LCM ? [SSC]
(a) 24 (b) 48 (c) 56 (d) 60
19. Two numbers, both greater than 29, have HCF 29 and LCM 4147. The sum of the numbers is [SSC]
(a) 666 (b) 669 (c) 696 (d) 966
20. The least number which when divided by 5, 6, 7 and 8 leaves 3 as remainder, but when divided by 9 leaves no remainder, is [LIC AAO]
(a) 1677 (b) 1683 (c) 2523 (d) 3363

21. The least multiple of 7, which leaves a remainder of 4, when divided by 6, 9, 15 and 18 is [LIC AAO]
(a) 74 (b) 94 (c) 184 (d) 364
22. The least number which when divided by 12, 15, 20 and 54 leaves in each case a remainder of 8, is [RRB]
(a) 504 (b) 536 (c) 544 (d) 548
23. The smallest number which when diminished by 7, is divisible by 12, 16, 18, 21 and 28 is [LIC]
(a) 1008 (b) 1015 (c) 1022 (d) 1032
24. What will be the least number which when doubled will be exactly divisible by 12, 18, 21 and 30?
(a) 196 (b) 630 (c) 1260 (d) 2520
25. The greatest number which on dividing 1657 and 2037 leaves remainders 6 and 5 respectively is [RRB]
(a) 123 (b) 127 (c) 235 (d) 305
26. Let N be the greatest number that will divide 1305, 4665 and 6905, leaving the same remainder in each case. Then sum of the digits in N is [SSC]
(a) 4 (b) 5 (c) 6 (d) 8
27. Find the greatest number that will divide 43, 91 and 183 so as to leave the same remainder in each case. [LIC]
(a) 4 (b) 7 (c) 9 (d) 13
28. The maximum number of students among them 1001 pens and 910 pencils can be distributed in such a way that each student gets the same number of pens and same number of pencils is [SSC]
(a) 91 (b) 910 (c) 1001 (d) 1911
29. The greatest possible length which can be used to measure exactly the lengths 7 m, 3 m 85 cm, 12 m 95 cm is [RRB]
(a) 15 cm (b) 25 cm (c) 35 cm (d) 42 cm
30. The HCF and LCM of two numbers are 11 and 385 respectively. If one number lies between 75 and 125, then that number is [CBI]
(a) 77 (b) 88 (c) 99 (d) 110
31. The HCF of two numbers is 23 and the other two factors of their LCM are 13 and 14. Then larger of the two number is [SSC]
(a) 276 (b) 299 (c) 322 (d) 345
32. Product of two co-prime numbers is 117. Their LCM should be [CBI]
(a) 1 (b) 117
(c) equal to their HCF (d) cannot be calculated
33. If the sum of two numbers is 55 and the HCF and LCM of these numbers are 5 and 120 respectively, then the sum of the reciprocals of the numbers is equal to [CDS]
(a) $\frac{55}{601}$ (b) $\frac{601}{55}$ (c) $\frac{11}{120}$ (d) $\frac{120}{11}$
34. The LCM of two numbers is 495 and their HCF is 5. If the sum of the numbers is 100, then their difference is [SSC]
(a) 10 (b) 46 (c) 70 (d) 90
35. The HCF of two numbers is 11 and their LCM is 7700. If one of the numbers is 275, then the other is [Section Officer]
(a) 279 (b) 283 (c) 308 (d) 318

36. The product of two numbers is 2028 and their HCF is 13. The number of such pairs is [CBI]
 (a) 1 (b) 2 (c) 3 (d) 4
37. The sum of two numbers is 528 and their HCF is 33. The number of pairs of numbers satisfying the above conditions is [CBI]
 (a) 4 (b) 6 (c) 8 (d) 12
38. Three numbers are in the ratio 1 : 2 : 3 and their HCF is 12. The numbers are [Section Officer]
 (a) 4, 8, 12 (b) 5, 10, 15 (c) 10, 20, 30 (d) 12, 24, 36
39. The LCM of 22, 54, 108, 135 and 198 is [MBA]
 (a) 330 (b) 1980 (c) 5940 (d) 11880
40. HCF of $4 \times 27 \times 3125$, $8 \times 9 \times 25 \times 7$ and $16 \times 81 \times 5 \times 11 \times 49$ is [CBI]
 (a) 180 (b) 360 (c) 540 (d) 1260
41. $\frac{1095}{1168}$ when expressed in simplest form is [MBA]
 (a) $\frac{13}{16}$ (b) $\frac{15}{16}$ (c) $\frac{17}{26}$ (d) $\frac{25}{26}$
42. Three numbers which are co-prime to each other are such that the product of the first two is 551 and that of the last two is 1073. The sum of the three number is [SSC]
 (a) 75 (b) 81 (c) 85 (d) 89
43. The LCM of two numbers is 48. The numbers are in the ratio 2 : 3. The sum of the numbers is [SSC]
 (a) 28 (b) 32 (c) 40 (d) 64
44. Three numbers are in the ratio 3 : 4 : 5 and their LCM is 2400. Their HCF is [MBA]
 (a) 40 (b) 80 (c) 120 (d) 200
45. The LCM of two prime numbers x and y ($x > y$) is 161. The value of $3y - x$ is [SSC]
 (a) -2 (b) -1 (c) 1 (d) 2
46. The least number, which when divided by 48, 60, 72, 108 and 140 leaves 38, 50, 62, 98 and 130 as remainders respectively is [CBI]
 (a) 11115 (b) 15110 (c) 15120 (d) 15210
47. Let the least number of six digits, which when divided by 4, 6, 10 and 15, leave in each case the same remainder of 2, be N . The sum of the digits in N is [SSC]
 (a) 3 (b) 4 (c) 5 (d) 6
48. Find the smallest number of five digits exactly divisible by 12, 16, 18 and 27.
 (a) 10432 (b) 10342 (c) 10368 (d) 10638
49. Find the largest number of four digits exactly divisible by 14, 24, 27 and 32.
 (a) 9968 (b) 9048 (c) 6048 (d) 6408
50. The LCM of $2^2 \times 3^2 \times 5 \times 11$, $2^4 \times 3^4 \times 5^2 \times 7$ and $2^5 \times 3^3 \times 5^2 \times 7^2 \times 11$ is
 (a) $2^3 \times 3^2 \times 5$ (b) $2^5 \times 3^4 \times 5^2$
 (c) $2^3 \times 3^2 \times 5 \times 7 \times 11$ (d) $2^5 \times 3^4 \times 5^3 \times 7^2 \times 11$



1. (e) 2. (i) (d), (ii) (c) 3. (d) 4. (d) 5. (b) 6. (b) 7. (b) 8. (b) 9. (b)
 10. (c) 11. (a) 12. (d) 13. (c) 14. (c) 15. (d) 16. (c) 17. (c) 18. (d) 19. (c)